Assignment - 0

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Q-1) (a) $\{1, 2, 1, 2, 3\}$

Z-transform :- $X(z) = 1 + 2z^{-1} + 1z^{-2} + 2z^{-3} + 3z^{-4}$

ROC : Entire Z-plane.

Causal : Yes (Start from $n = 0$)

Stable : Yes

(b) $x(n) = (n+1)^2 u(n)$

Z-transform $X(z) = \sum_{n=0}^{\infty} (n+1)^2 z^{-n}$.

ROC: $|z| > 1$

Causal : Yes

Stable : No

(c) $x(n) = \log(n)$

Z-transform : Not defined at $n = 0$

Causal : No

Stable : No

d) $x(n) = 2^n u(n) - 0.5^n u(-n-1)$

Z trans. : $X(z) = \frac{1}{1-2z^{-1}}, \quad |z| > 2$ (Right)

$+ \frac{z}{z-0.5}, \quad |z| < 0.5$ (left)

ROC : No. overlapping ROC → Z-transform undefined

Causal : No

Stable : No

e) $x(n) = u(n-6)$

This is 1 for $n \leq -6$

$X(z) = \sum_{n=-\infty}^{-6} z^{-n} = \frac{z^6}{1-z}, \quad |z| < 1$

f) $x(n) = \delta(n+5)$

z - trans $X(z) = z^5$

ROC = Entire z-plane

causal : Yes

stable : Yes

Q-2) Inverse z-transform

(a) $H(z) = 1 + z^{-1}$

$h(n) = \delta(n) + \delta(n-1)$

b) $H(z) = \log(1+z^{-1})$

$h(n) = \frac{(-1)^{n+1}}{n} \quad n \geq 1$

c) $H(z) = z \cdot e^{\wedge}(2z^{-3})$

$e^{2z^{-3}} = \sum_{k=0}^{\infty} \frac{(2z^{-3})^k}{k!}$, multiply by z

d) $H(z) = 7/(1-4z^{-1})(1-2z^{-1}))$

$H(z) = \frac{A}{1-4z^{-1}} + \frac{B}{1-2z^{-1}} \Rightarrow h(n) = A \cdot 4^n + B \cdot 2^n, n \geq 0$

e) $H(z) = 1/((z-3)^n n(-1))$

This is undefined.

Q-3) $y(n) = y(n-1) + 5x(n) + 2x(n-1)$

Impulse Response $h(0) = 5$, $h(1) = h(0) + 2 = 7, \ldots$

frequency Response (z-domain):

$H(z) = \frac{5+2z^{-1}}{1-z^{-1}} \Rightarrow H(e^{jw}) = \frac{5+2e^{-jw}}{1-e^{-jw}}$

magnitude and phase : use magnitude and angle formulas

Output for $x(n) = 2^n u(n)$: $Y(z) = H(z) \cdot X(z)$, then inverse z-transform.

Q-4) $H(z) = 4/(z^2-3)$

$$H(z) = \frac{4}{(z-\sqrt{3})(z+\sqrt{3})}$$

Partial fraction → inverse z-transform gives $h(n)$
Unit step response : convolve $h(n)$ with $u(n)$

Q-5) convolution via z-Transform

$x(n) = 2^n u(n) \Rightarrow X(z) = \frac{1}{1-2z^{-1}}, \ |z| > 2$

$h(n) = 5^n u(n-1) \Rightarrow H(z) = \frac{z}{z-5}, \ |z| < 5$

No overlapping ROC → use time domain convolution

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Q-6) DTFTs

(a) $(n+1)a^n u(n)$

$$X(e^{j\omega}) = \frac{1-ae^{-j\omega}}{(1-ae^{-j\omega})^2}, \ |a|<1$$

(b) $\sin(\omega n)/\pi n$

sinc function → Rectangular frequency response

c) $\cos(\omega n + \beta)$

DTFT contains two impulses at $\pm\omega$.

d) $e^{(j\omega n)} \cdot u(n)$

$$X(e^{j\Omega}) = \frac{1}{1-e^{j(\Omega-\omega)}}$$

Q-7 IDTFT

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

Q-8) Linear & circular convolution:
Compute
Linear: by time domain summation.
Circular: by wrapping modulo n.

Q-9) $|H(e^{jw})| = 1$
use phase to find
$h(n) = \delta(n-k)$, shifted impulse.

output for $x(n) = \cos(\pi n)u(n)$
$y(n) = \cos(\pi(n-k))u(n-k)$

Q-10) properties of z-transform

1. Differentiation: $nx(n) \leftrightarrow -z \frac{dX(z)}{dz}$

2. Time shifting: $x(n-k) \leftrightarrow z^{-k}X(z)$

3. convolution: $x(n) * h(n) \leftrightarrow X(z)H(z)$

4. Time Reversal: $x(-n) \leftrightarrow X(z^{-1})$

Q-11) If $x(n)$ is real then:

$X(e^{jw}) = X^*(e^{-jw})$
Even magnitude odd phase

Q-12) $x(n) = \{1,1,2,2\}$ $h(n) = \{1,2,3,4\}$
(a) linear from circular: zero pad to $N=7$
use circular conv
b) circular from linear: wrap circular summing method
c) Linear from DFT / IDFT:
Pad both to length $\geq n+m-1$
compute DFT → multiply → IDFT

Q-13) 8-point DFT of $x(n) = \cos(n\pi/4)$

$x(n) = \cos(n\pi/4) \rightarrow$ periodic with $N = 8$

DFT : Peaks at bins 1 and 7

Amplitude spectrum : 1 at $k = 1$ and 7

Phase : 0 at $k = 1$, π at $k = 7$

Q-14) System Properties Table,

System	Stable	Causal	Linear	Time-Invariant	Memory
$y(n) = x(n-n_0)$	✓	✓	✓	✓	✓
$y(n) = x(-n)$	✓	✗	✓	✓	✓
$y(n) = x^2(n)$	✓	✓	✗	✓	✓
$y(n) = \log(x(n))$	✓	✓	✗	✓	✓
$y(t) = dx(t)/dt$	✗	✓	✓	✓	✗

Q-15) Plot $x(n) = u(n) + 3\delta(n-1) + 4u(-n-3)$

$x(n)$: Step + impulse + flipped step

$x(n-1)$: Shift Right

$x(2n)$: Compression

$x(n/2)$: Expansion

$x^2(n)$: Square each value

$x(-n)$: Time reversal

Q-16) Periodicity

Signal	Periodic	Period
$e^{j\pi/6}$	✓	12
$\sin(3n)$	✓	Approximate
$e^{jn/1.41}$	✗	Aperiodic
$\sin(3\pi n) + \cos(6\pi n)$	✓	2
$\frac{\sin(\pi n)}{\pi n}$	✗	Aperiodic